

Does the Standard Sparse-Autoencoder Feature-Absorption Metric Over-Report for Non-Spelling Features?

A Cross-Metric, Cross-Model Study

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Abstract

Sparse autoencoders (SAEs) are meant to decompose language-model activations into monosemantic features, but *feature absorption*—where a general feature silently fails on some tokens while a token-specific latent carries the concept—undermines that promise. Absorption is quantified in the literature almost exclusively on a single task, “does this token start with letter X ,” and with two different metrics: an older *ablation* (behavioral) metric and the current SAEBench-standard *projection* (representational) metric. We ask whether the two metrics agree once absorption is measured on a structurally different, non-spelling property. Running both metrics on IS-CAPITALIZED across Gemma-2-2b and Gemma-2-9b (Gemma Scope 16k SAEs), we find that they *disagree*: the projection metric reports single-dominant-latent absorption modestly above its random-direction null (0.14, 95% CI [0.085, 0.200] vs. 0.04; $n=130$ —marginal), whereas the ablation metric reports exactly zero on 273/273 candidate tokens, on *both* models. Two controls rule out the mundane explanations: it is not task performance (Gemma-2-9b performs IS-CAPITALIZED at 0.91 accuracy, yet ablation stays 0.0) and not a dead layer (at the same layer, spelling ablation fires at 0.045; at matched task accuracy it fires at ~ 0.026 for spelling while IS-CAPITALIZED remains 0.0). We conclude that the field-standard projection absorption metric appears to over-report single-latent absorption for a distributed structural feature that has no causal correlate. As a secondary result, we show that ablation-based absorption is itself task-accuracy-sensitive within the spelling task. We reproduce the known first-letter absorption and its L0 dependence as pipeline validation. Effects are near-floor and the study spans one property, one model family, and one SAE recipe; we treat the result as a metric-validity caveat and an argument that representational absorption signals require causal validation.

1 Introduction

Sparse autoencoders (SAEs) have become a leading tool for decomposing the internal activations of language models into a large, sparse set of features that are, ideally, monosemantic and human-interpretable [1, 5, 6, 11]. A central threat to this interpretation is *feature absorption* [3]: a latent that appears to encode a general concept (e.g. “starts with the letter S”) silently deactivates on a subset of tokens, while a more specific, token-aligned latent (e.g. a “short” latent) absorbs the general direction and carries the concept for those tokens. Absorption means the “general” feature is not what it seems, which is corrosive for any downstream use of SAE features as reliable concept detectors.

Despite its importance, absorption has been measured almost exclusively on one family of tasks: first-letter (and, in benchmarks, first-letter only) spelling, where ground truth is trivially available [3, 9]. Chanin et al. explicitly list “examples of feature absorption unrelated to

character identification” as open future work. Moreover, two distinct metrics are in use: (i) an *ablation* (behavioral) metric that attributes the model’s task answer to individual SAE latents via integrated gradients [3], and (ii) a *projection* (representational) metric that decomposes the residual stream’s projection onto a concept probe direction into per-latent contributions. The projection metric is the current standard adopted by SAEBench [9], specifically because the ablation effect decays toward zero in later layers.

This raises a simple, previously untested question: **do the two absorption metrics agree once we move off the spelling task?** We answer it for structural, non-spelling token properties—primarily IS-CAPITALIZED (whether a token begins with an uppercase letter)—and find that they do not.

Contributions.

1. We run *both* standard absorption metrics (projection and ablation) on structural token properties, contrasting them directly. To our knowledge this specific comparison has not been reported. (We do *not* claim the first measurement on non-character properties: Chanin et al. [4] already probe part-of-speech with k -sparse probing.)
2. We report a *metric disagreement*: for IS-CAPITALIZED, projection reports above-null absorption while ablation reports a clean zero, with bootstrap confidence intervals.
3. We isolate the cause with two controls—a cross-model task-ability control (Gemma-2-9b) and a layer-/accuracy-matched spelling control—ruling out both “the model cannot do the task” and “the layer is dead.” The residual explanation is a genuine property-level difference in absorption structure.
4. As a secondary, controlled result, we show the ablation metric is itself sensitive to task accuracy within the spelling task, whereas the projection metric is not.

2 Background and Related Work

Sparse autoencoders and monosemanticity. An SAE reconstructs a model activation $\mathbf{x} \in \mathbb{R}^d$ as $\hat{\mathbf{x}} = W_{\text{dec}} \mathbf{a} + \mathbf{b}$ with sparse codes $\mathbf{a} = \text{enc}(\mathbf{x}) \geq 0$, trained to trade off reconstruction against sparsity [1, 5]. We use Gemma Scope [10], the open suite of JumpReLU SAEs trained on Gemma-2 [7].

Feature absorption and its two metrics. Chanin et al. [3] define absorption and measure it on the first-letter task. Their metric is causal: on a token where the general (“main”) latent is silent, they use integrated-gradient attribution to check whether a single latent, aligned with a linear probe for the concept, most strongly supports the model’s answer. SAEBench [9] standardizes a *projection* variant—decomposing the residual’s projection onto the probe direction into per-latent contributions— noting that the ablation signal “goes to zero in later layers.” Thus the ablation metric is effectively deprecated in favor of projection; a corollary is that “ablation reads 0” is, in later layers, a known artifact rather than evidence of no absorption. Our contribution rests precisely on distinguishing that artifact from a genuine zero via layer- and accuracy-matched controls.

Related mechanisms and mitigations. Chanin et al. [4] describe *feature hedging*, whereby narrow SAEs merge correlated features into distributed, partly-polysemantic latents, and already probe part-of-speech. Hedging is the most likely mundane mechanism for a structural concept being “represented but distributed,” and we engage it in the discussion. Matryoshka SAEs [2] are an architecture proposed to reduce absorption. Separately, Kantamneni et al. [8] find SAE

latents do not beat raw-activation probing baselines, situating our missing downstream “so what” in a broader debate about SAE utility.

3 Methods

Models and SAEs. We use Gemma-2-2b (residual stream, layer 12) and Gemma-2-9b (layer 20), each with the Gemma Scope JumpReLU residual SAE of width 16k, at an average $L0$ near 70–82. Layers are chosen well below the depth at which the ablation metric is known to fail. All activations are read at the word-token position of an in-context prompt.

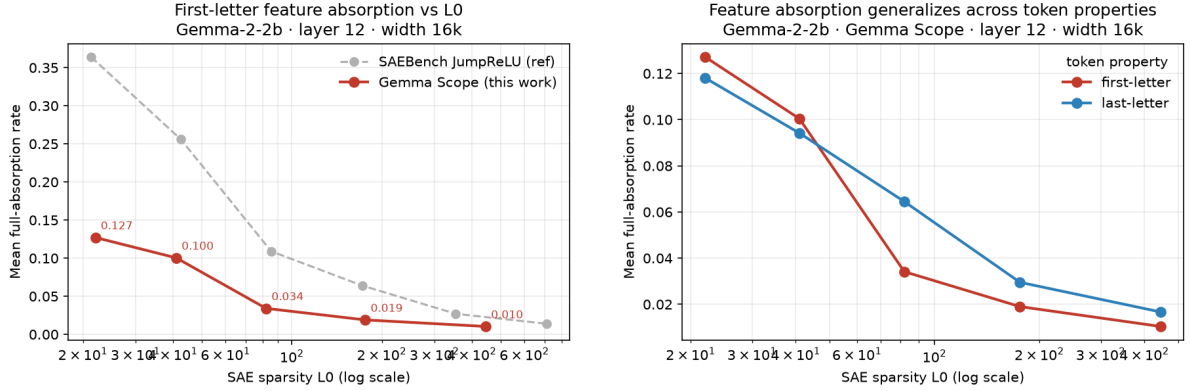
Token properties. The concept is elicited by an in-context task with an objective, tokenizer-derivable label: first-letter and last-letter (26-way), and the binary structural properties IS-CAPITALIZED, ALL-CAPS, and suffix--ING/-ED. Only IS-CAPITALIZED yields a well-powered candidate pool ($n=130\text{--}273$); the others are underpowered ($n=9\text{--}33$) and reported as anecdotes.

Pipeline. For each concept we (i) train a linear probe on the residual stream (the ground-truth concept direction); (ii) run k -sparse probing to identify the concept’s “main” SAE latent(s) via a feature-splitting F1 criterion; and (iii) collect *candidate false negatives*: tokens where the probe fires but the main latents do not. Absorption is then quantified on these candidates by two metrics.

Ablation (behavioral) metric. Following Chanin et al. [3], on each candidate we run integrated-gradient attribution of SAE latents to a logit-difference metric for the task answer; a token counts as absorbed if the main latents are silent, a single latent is the clear most-causal one, and it is probe-aligned. We report the absorption rate (absorbed / probe-true-positives), with an additive dominance threshold.

Projection (representational) metric. We decompose the residual’s projection onto the (unit) probe direction $\hat{\mathbf{p}}$ into per-latent contributions $c_i = a_i (W_{\text{dec},i} \cdot \hat{\mathbf{p}})$. On a candidate where the main latents are silent, absorption is present if one non-main latent dominates: $c_{(1)} \geq R c_{(2)}$ for a scale-free dominance ratio R . Crucially we add a **random-direction null**: the same statistic computed against random unit directions in place of $\hat{\mathbf{p}}$, averaged over five draws, to control for the fact that SAE sparsity makes *any* projection tend to be dominated by one active latent. We report the *conditioned* rate (fraction of main-silent candidates showing single-dominant absorption) so that the estimate does not bake in main-latent quality.

Task-accuracy measurement and controls. For each (property, model) we measure the model’s in-context accuracy on the task. To separate “ablation reads 0 because the concept is not causally absorbed” from the two mundane confounds, we use: (a) a **cross-model** control (Gemma-2-9b, which performs IS-CAPITALIZED far better than 2b); (b) a **layer-matched** spelling control (first-letter absorption at the same 9b layer/ $L0$); and (c) a **matched-accuracy** control, obtained by sweeping first-letter task accuracy through the number of in-context examples and by corrupting a fraction of in-context answers, so we can compare structural and spelling absorption at equal task accuracy. All statistics use bootstrap 95% confidence intervals; the single-class (binary-property) pipeline is a re-implementation of the 26-way one and was independently review-verified.



(a) Absorption vs. L_0 (Gemma Scope vs. SAEBench reference).

(b) First- vs. last-letter absorption vs. L_0 .

Figure 1: Reproduction. (a) The first-letter absorption– L_0 law is recovered on current Gemma Scope SAEs. (b) Last-letter tracks first-letter, so within spelling the phenomenon is not first-character-specific.

Table 1: Mean absorption rate vs. SAE sparsity L_0 (Gemma-2-2b, layer 12, width 16k). Both spelling properties show the expected monotone decrease with L_0 .

L_0	22	41	82	176	445
first-letter	0.127	0.100	0.034	0.019	0.010
last-letter	0.118	0.094	0.065	0.030	0.017

4 Results

4.1 Reproduction and the L_0 law

As pipeline validation we reproduce first-letter absorption on current Gemma Scope SAEs and recover the known dependence on SAE sparsity: absorption rises steeply as L_0 falls (Figure 1a, Table 1), sitting a consistent factor below SAEBench’s own JumpReLU numbers, as expected for the more thoroughly trained Gemma Scope SAEs. Last-letter behaves like first-letter (Figure 1b), confirming that within the spelling family absorption is not specific to the first character.

4.2 The two metrics disagree on is-capitalized

Table 2 and Figure 2 report the projection metric with its random-direction null. For spelling, projection absorption is large and far above the null (0.54 vs. 0.04 at $R=3$). For IS-CAPITALIZED, projection absorption is above the null but *marginal* at $n=130$ (0.14, CI [0.085, 0.200] vs. null 0.04, CI [0.008, 0.077]), and roughly $4\times$ weaker than spelling.

In sharp contrast, the *ablation* metric reports IS-CAPITALIZED absorption of exactly 0.0—on both Gemma-2-2b and Gemma-2-9b, and on all 273 candidate tokens of the 9b run (binomial 95% CI [0, 0.011]). The two metrics thus disagree: projection sees a (weak) absorber; ablation sees none.

4.3 Controls: not task-validity, not a dead layer

Table 3 and Figure 3 present the cross-model controls. Because the ablation metric is known to fail in later layers, the decisive question is whether the IS-CAPITALIZED zero is genuine. It is: (i) Gemma-2-9b performs IS-CAPITALIZED at 0.91 accuracy—far above 2b’s 0.60—yet ablation

Table 2: Projection (representational) single-dominant-latent absorption at dominance $R=3$, conditioned on the main-latent-silent candidate pool, with bootstrap 95% CIs, vs. a random-direction null.

property	projection [95% CI]	random null [95% CI]	n
first-letter	0.54 [0.51, 0.56]	0.04 [0.03, 0.05]	2085
IS-CAPITALIZED	0.14 [0.085, 0.200]	0.04 [0.008, 0.077]	130

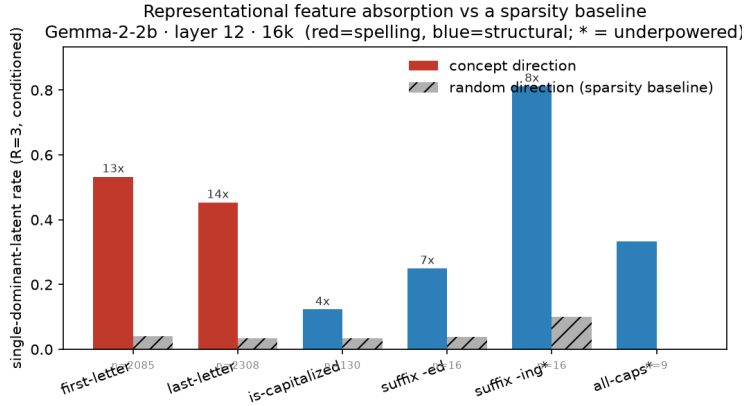


Figure 2: Projection absorption vs. a random-direction sparsity baseline (strict dominance $R=3$). Spelling (red) is 13–14× above the null; structural properties (blue) are above the null but much weaker, and ALL-CAPS/suffix are underpowered (*).

absorption remains 0.0; and (ii) at the *same* 9b layer/ $L0$, first-letter ablation absorption is 0.045, so the metric demonstrably fires there. Finally, using the accuracy sweep of §4.4, at matched task accuracy ~ 0.9 first-letter ablation absorption is ~ 0.026 while IS-CAPITALIZED is 0.0. The structural zero is therefore not explained by task ability or layer.

4.4 The ablation metric is task-accuracy-sensitive

Sweeping first-letter task accuracy from 0.53 to 0.97 (via the number of in-context examples and via corrupting a fraction of in-context answers) moves the *ablation* absorption rate monotonically from 0.011 to 0.030, while the *projection* rate stays flat at ~ 0.60 (Figure 4). Two independent accuracy manipulations agree. This is a genuine, if secondary, reliability caveat: the ablation metric conflates “the concept is causally absorbed” with “the model performs the eliciting task,” whereas the projection metric does not. It also implies that comparisons using the ablation metric must control for task accuracy—which our structural controls above do.

5 Discussion

The defensible reading is a *metric-validity* one. For a distributed structural feature (IS-CAPITALIZED), the SAE Bench-standard projection metric’s “single dominant latent” criterion reads modestly above chance, yet there is no causal single-latent correlate: ablation, in a regime where it works for spelling and where the model performs the task, reports a clean zero. In other words, the projection metric appears to *over-report* single-latent absorption for a feature whose concept direction is carried by the residual but not concentrated in one causally-load-bearing latent.

We cannot fully rule out the most natural alternative, and we state it plainly: *feature hedging* [4]. Capitalization is a high-frequency, broadly co-occurring concept; a 16k SAE may simply

Table 3: Cross-model behavioral (ablation) absorption. Spelling absorbs on both models; IS-CAPITALIZED on neither, even as its task accuracy rises from 0.60 (2b) to 0.91 (9b).

property	model / layer	task accuracy	ablation absorption
first-letter	Gemma-2-2b / L12	~ 0.95	0.032
first-letter	Gemma-2-9b / L20	~ 0.96	0.045
IS-CAPITALIZED	Gemma-2-2b / L12	0.60	0.000
IS-CAPITALIZED	Gemma-2-9b / L20	0.91	0.000

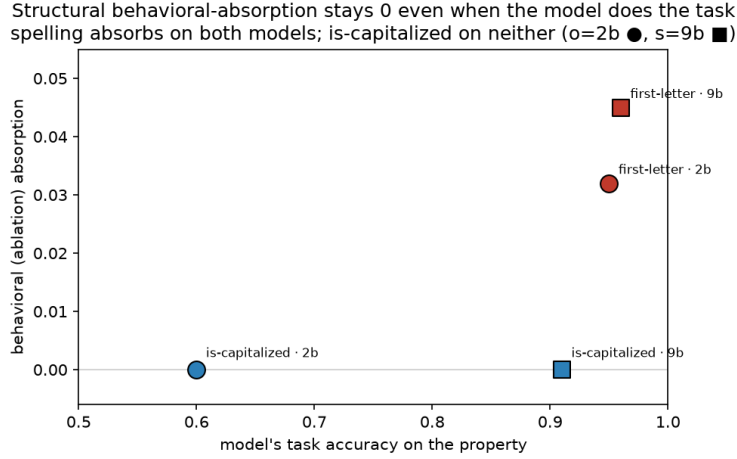


Figure 3: The structural (blue) behavioral-absorption zero persists even when the model’s task accuracy rises from 0.60 to 0.91, while spelling (red) absorbs on both models—so the zero is not a task-performance artifact.

spread it across several latents, so that “one dominant latent by projection geometry” is a weak, partly-geometric signal rather than a token-aligned absorber the model uses. Under this view our result is best read not as a deep “representational-vs-causal dissociation” but as concrete evidence that a representational absorption signal, at least for distributed features, does not imply a causal one—consistent with the broader argument that interpretability claims need causal validation.

6 Limitations

Our effects are small. Spelling behavioral absorption is only ~ 0.01 – 0.05 anywhere, and the IS-CAPITALIZED projection signal is marginal above its null at $n=130$. Only one structural property is well-powered (ALL-CAPS and the suffix properties have $n=9$ – 33 and are anecdotal), so “property-type-dependent” is effectively an $n=1$ claim. Both models are from one family (Gemma-2, cross-scale only) and we use a single SAE recipe (JumpReLU/Gemma Scope); we run no non-JumpReLU or Matryoshka SAE. Finally, we show no downstream task in which trusting the projection-flagged absorption would demonstrably mislead—the “so what” that would elevate a caveat into an actionable finding.

7 Conclusion

Measuring SAE feature absorption off the spelling task exposes a disagreement between its two standard metrics: for IS-CAPITALIZED, the projection metric reports weak-but-above-null

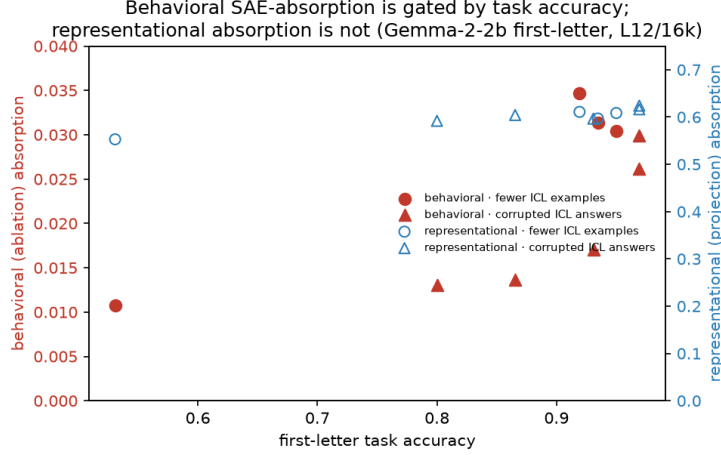


Figure 4: Within first-letter, behavioral (ablation) absorption (red) rises with task accuracy under two independent manipulations, while representational (projection) absorption (blue) is flat.

absorption that the ablation metric denies, and controls show the denial is neither a task-performance nor a layer artifact. The most likely mechanism is feature hedging, and the honest headline is a metric-validity caveat: representational absorption signals for distributed structural features can lack a causal correlate. The natural next steps that would turn this from a single marginal case into a robust, transferable claim are to establish the disagreement on a second SAE architecture (e.g. TopK or Matryoshka) and a second, well-powered structural property, and to exhibit a downstream decision that the mis-measurement would flip.

Reproducibility. All code, figures, and per-run numbers behind every claim above are available in the accompanying repository, including the model-agnostic loaders used for the cross-model controls and the independently review-verified single-class absorption pipeline.

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